

Junyu Liao

Selected Works 2020-2025
B.Arch Guangzhou University

CONTENT

PORLOGUE

Architecture is the carrier of humanities, social activities, and traditional culture. Thus, integrating with the environment culture and adaptedto future social life is what I take consider with in my designs. Also, the works of **Louis Kahn** made me see a bigger picture of design, inspiring me to paying more attention to the integration of architecture with vernacular context, environment and urban elements. Inspired by **Jane Jacobs** with her option about people being the main body of cities, human activities start to become important factors and the core concept in my projects.

01 Urban Acupressure Points

Complex Spatial Revitalization and
Renovation of Old Urban Areas

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Ideal Future Residential Ensemble

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Group work

15-20

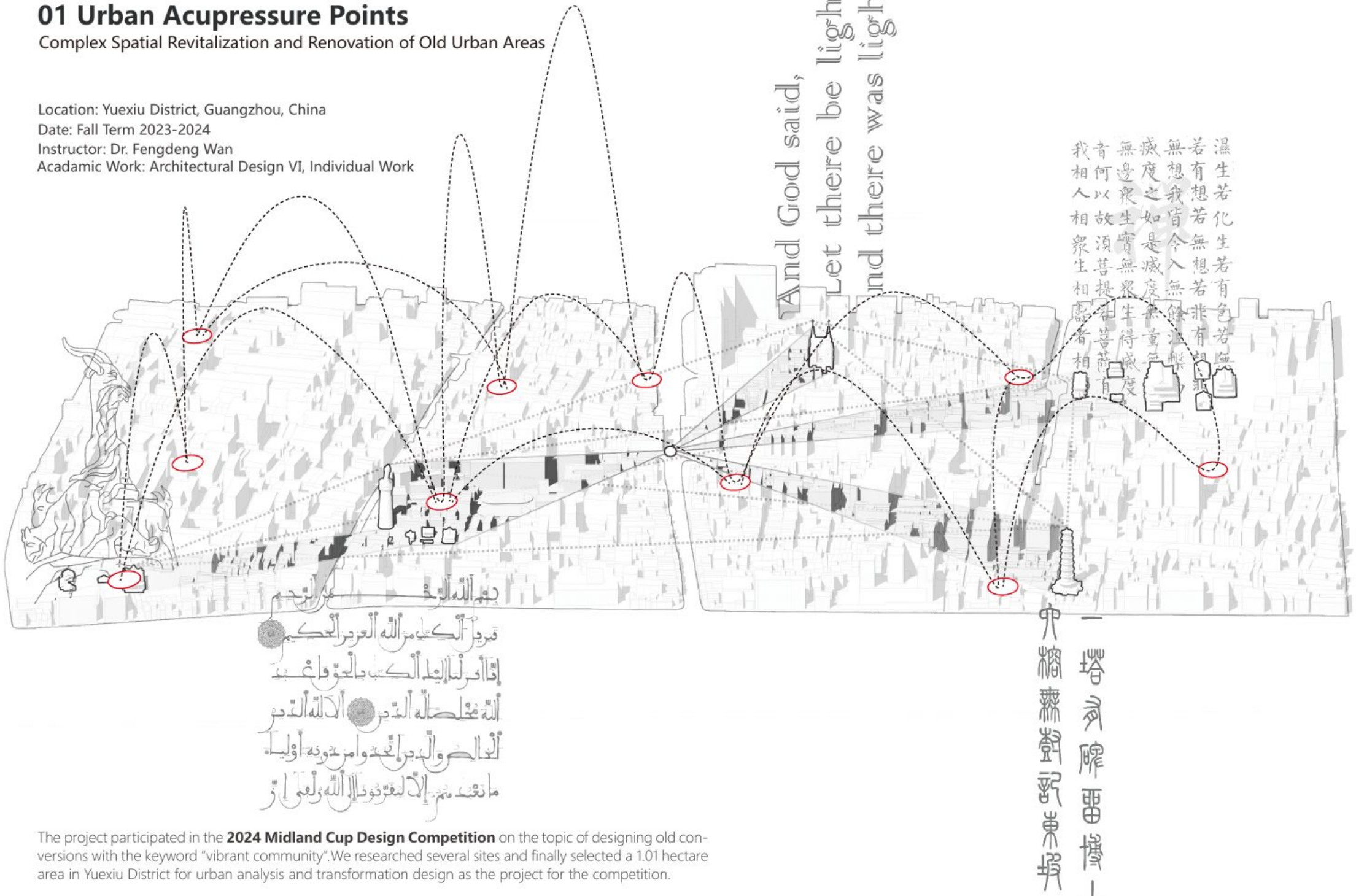
05 Other Works

21

01 Urban Acupressure Points

Complex Spatial Revitalization and Renovation of Old Urban Areas

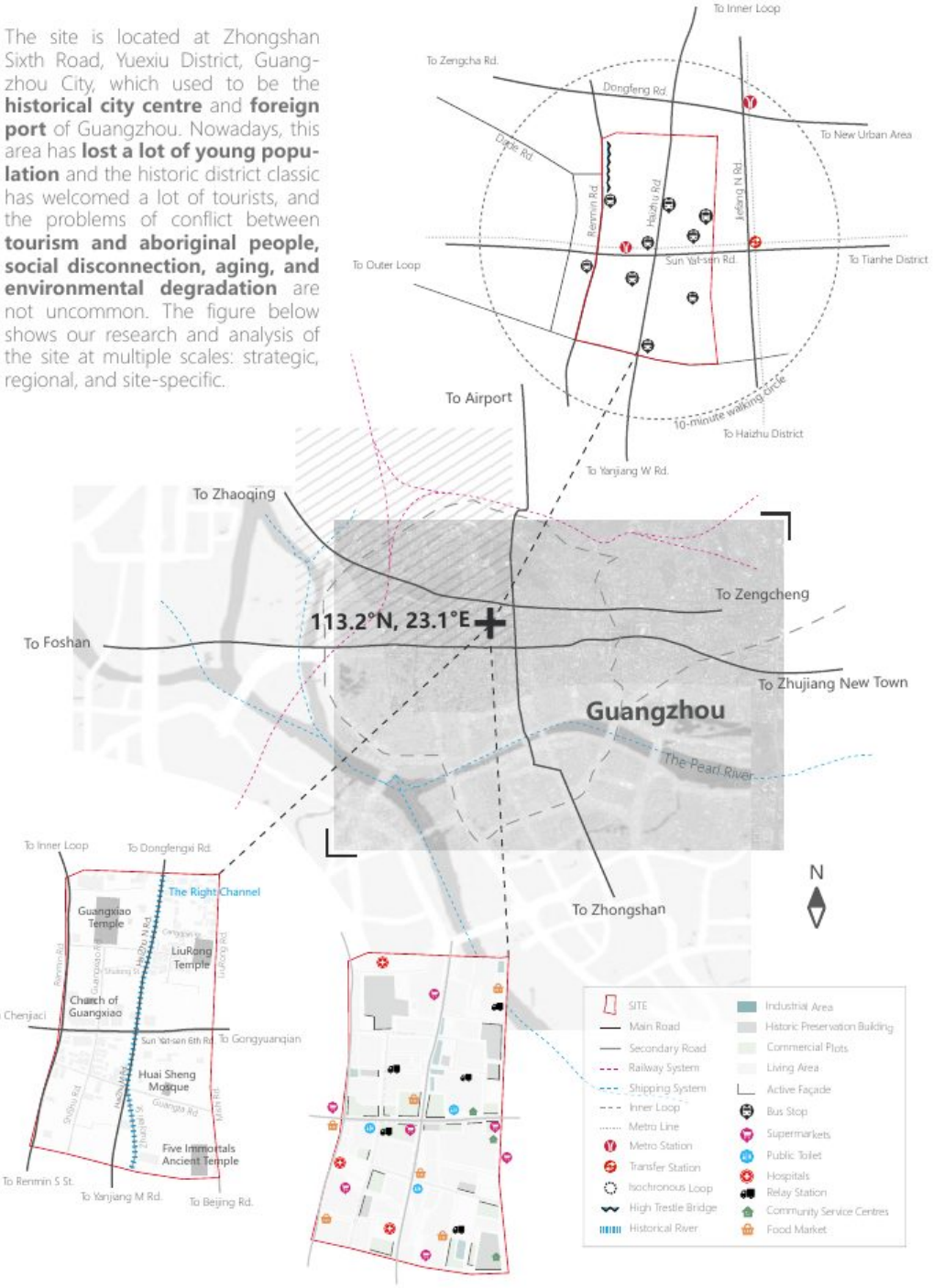
Location: Yuexiu District, Guangzhou, China
Date: Fall Term 2023-2024
Instructor: Dr. Fengdeng Wan
Acadamic Work: Architectural Design VI, Individual Work



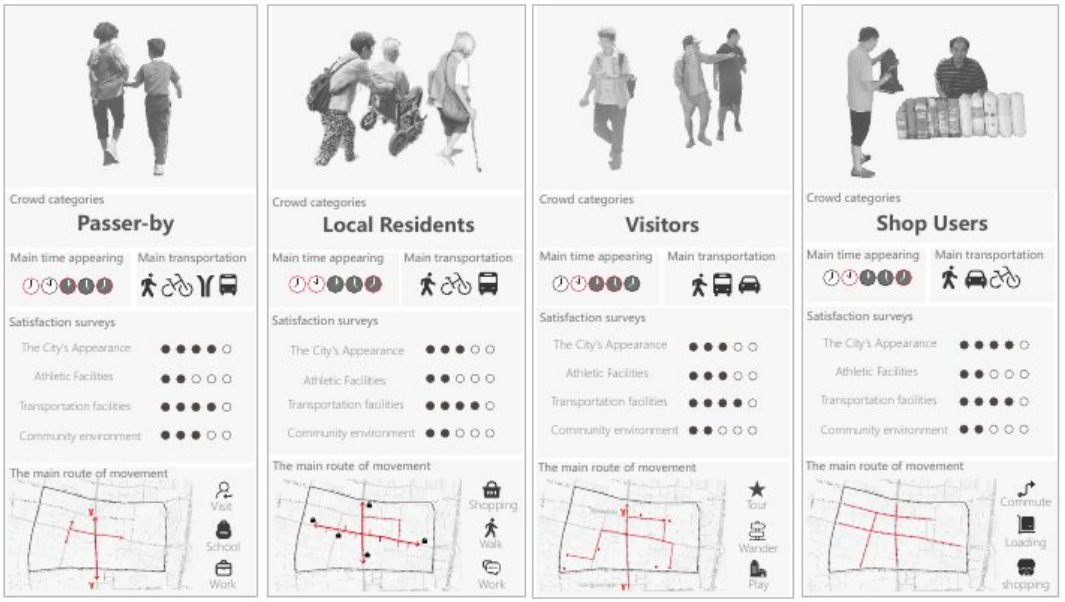
The project participated in the **2024 Midland Cup Design Competition** on the topic of designing old conversions with the keyword "vibrant community".We researched several sites and finally selected a 1.01 hectare area in Yuexiu District for urban analysis and transformation design as the project for the competition.

Site Analysis

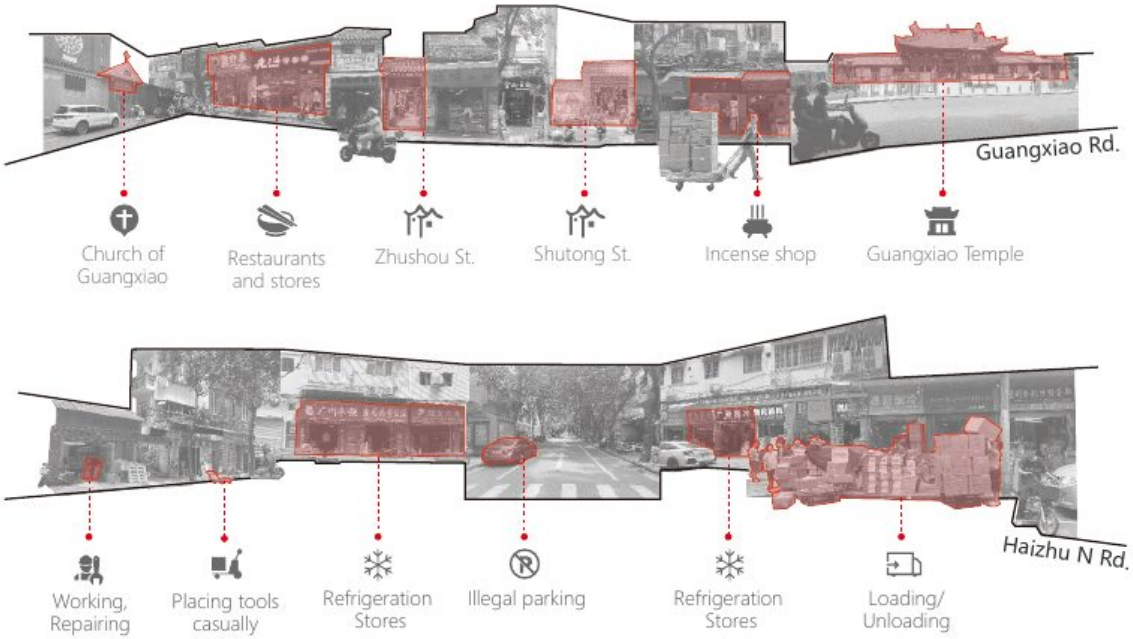
The site is located at Zhongshan Sixth Road, Yuexiu District, Guangzhou City, which used to be the **historical city centre** and **foreign port** of Guangzhou. Nowadays, this area has **lost a lot of young population** and the historic district classic has welcomed a lot of tourists, and the problems of conflict between **tourism and aboriginal people, social disconnection, aging, and environmental degradation** are not uncommon. The figure below shows our research and analysis of the site at multiple scales: strategic, regional, and site-specific.



Crowd Surveys



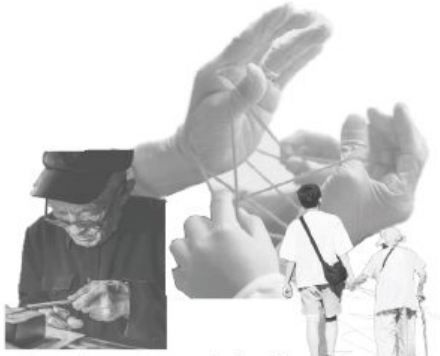
Block Composition



Logical Introduction



Reform and opening up, monetization of shanty-town reform, and a large number of young people flock to the city.



How to get young people back? On the basis of family affection, it is more important to establish production relations and tap the relationship between supply and demand.



With the development of e-commerce, under the policy of **the Belt and Road** Initiative, economic conditions permit, **Made in China — the national is the world.**

Urban employment has almost reached saturation, and phenomena such as **Empty Nesters, Left-behind Children** and **Silver-haired Addiction** are frequent.



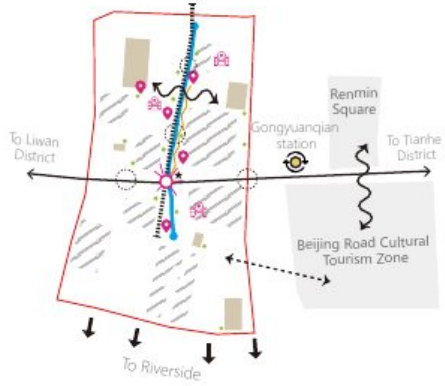
Through the cultural inheritance of **bringing the old with the new**, the intangible cultural heritage is transformed into modern product output.



A Missing Chapter

objective assessment and corresponding strategies

Design Considerations



Opportunities

- The site
- Historic Listed Area
- Developed Green Space
- Existing Culverts
- Potential Sports Squares
- Existing School's Playground
- Main Transport Hub
- Main Transport Connection
- Existing Strong Connection
- Potential Enhancing Connection
- Key View
- Historic Listed Trees

Constraints

- Potential Unsafe Interations
- Potential Unsafe blocks
- Dense living area
- Barrier
- High-rise Buildings

Objectives & Actions

A



Focusing on the feelings of Cantonese people about the **banyan tree**, the node installation is explored under the banyan tree, and the space under the tree is used.

B



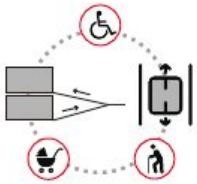
Let the old man with the best skills have the desire to pass it on, **provide employment for the young people**, and establish the production relationship between the two.

C



The surplus income obtained through operation can be used to **feed back** the initial cost of construction & subsequent maintenance, forming a virtuous circle.

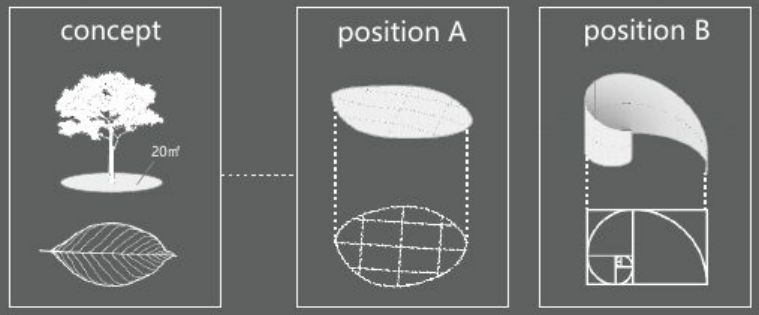
D



Considering the composition of the community, we should have a **humanistic spirits** and care for social workers like Sanitation workers, disabled people, stray cats and dogs, etc.

Community Activity System

About the organization of the nodes and the age-appropriate design of the block

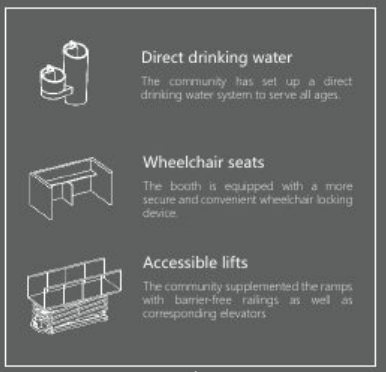


• Community recreation

The Leisure and Shelter nodes will provide a more comfortable roaming experience for the entire community.

• Sporty atmosphere

The sports nodes offer a variety of terrain that attracts neighborhood sports enthusiasts, where sports competitions can be held.



Direct drinking water

The community has set up a direct drinking water system to serve all ages.

Wheelchair seats

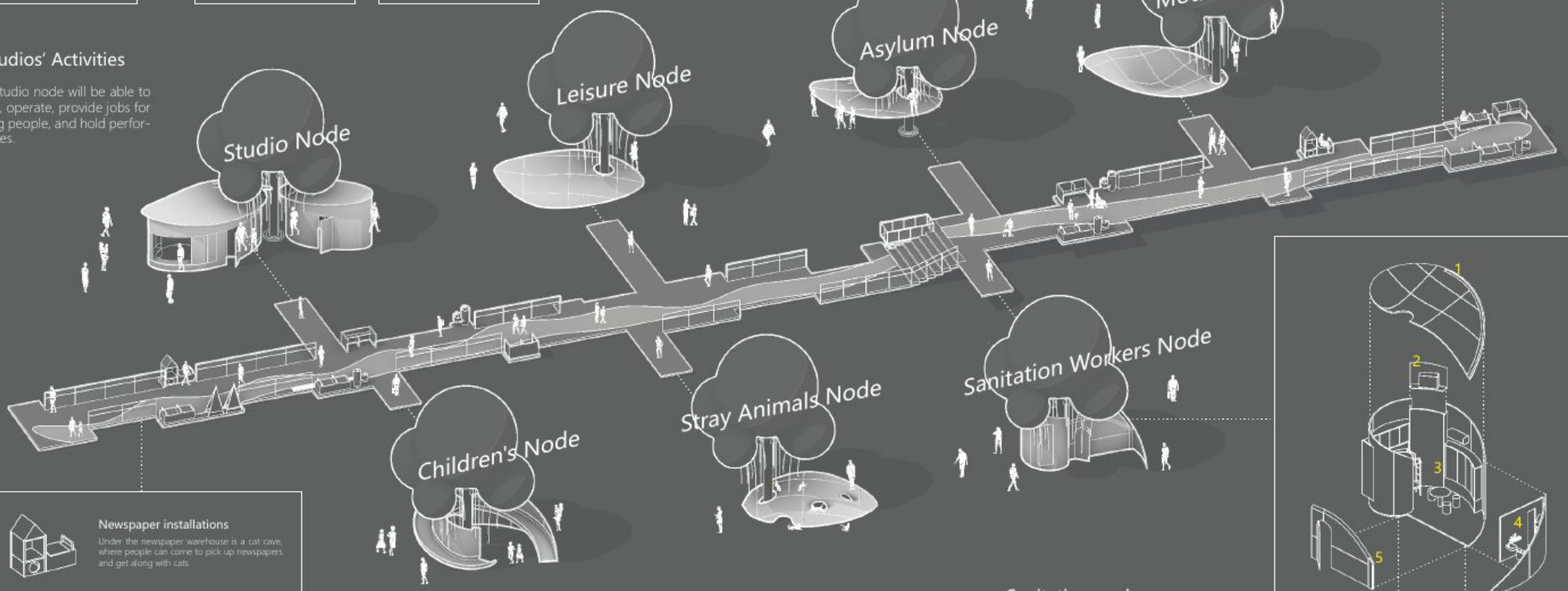
The booth is equipped with a more secure and convenient wheelchair locking device.

Accessible lifts

The community supplemented the ramps with barrier-free railings as well as corresponding elevators.

• Studios' Activities

The studio node will be able to teach, operate, provide jobs for young people, and hold performances.





Newspaper installations

Under the newspaper warehouse is a cat cave, where people can come to pick up newspapers and get along with cats.

Dog walking lounge

There are many people walking their dogs in the community, and there is also a short rest place for the dogs.

Age-friendly seats

The chairs are equipped with upholstery and armrests.

• Children's activities

Children can play here after school, and there are plenty of facilities and a variety of games.

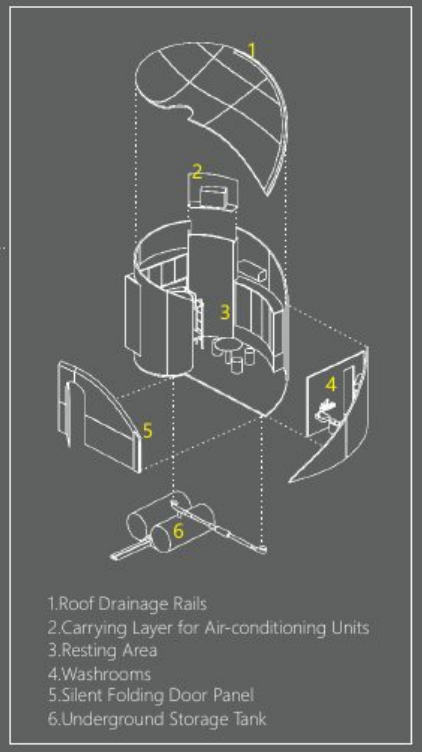
• Live stream interactions

The live broadcast can be connected to the feeding of stray cats and dogs, and the profits obtained can be fed back to the maintenance of the device to achieve a good operation cycle.



• Sanitation workers care

Dedicated nodes provide living conditions where hard-working community sanitation workers can take short breaks.



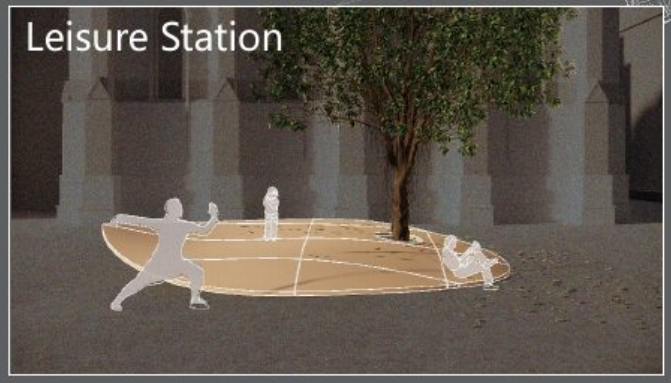
1.Roof Drainage Rails
2.Carrying Layer for Air-conditioning Units
3.Resting Area
4.Washrooms
5.Silent Folding Door Panel
6.Underground Storage Tank


Cultural Center


Workers' Rest Area


Children's Park


Tongning Alley Community
Church of Guangxiao
Zhushou Alley Community
Guangxiao Temple
Guangxiao Rd Community
Renmin Mid Community
Guangdong Zhiyong School
Guangzhou Trade Vocational High School
Liurong Temple
Waisheng Mosque
Fuquan Community
Progress Lane Community
Wuxianguan Daoist temple

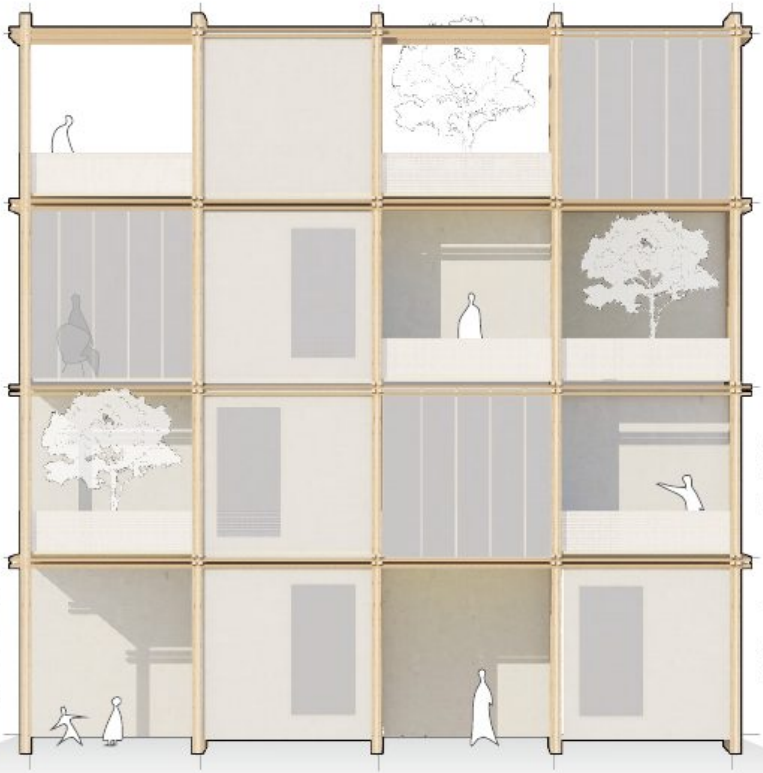

Leisure Station


Animal Room


Sports Corner

02 "TOD" in 2046

The Noise Problem leads to the Exploration of the Ideal Future Residential Ensemble



Location: Baiyun District, Guangzhou, China
Date: Spring Term 2023-2024
Instructor: Prof. Yang Zhao
Acadamic Work: Architectural Design V, Individual Work

A happened conversation learned that a friend lives in this neighborhood, gave me a lot of feedback on the noise problem, which I used as a subject to start the program.

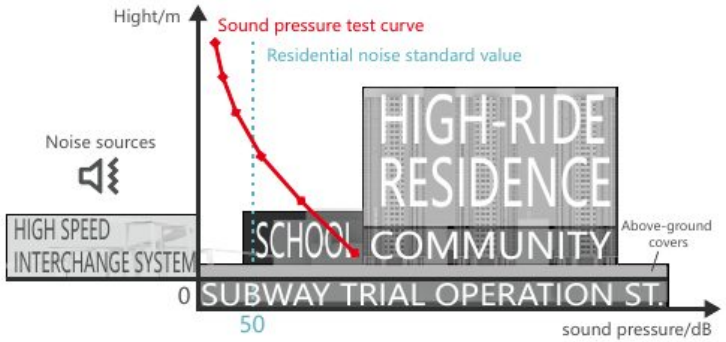
The project is located in Baiyun District, Tinggang Station subway superstructure real estate projects, I analyzed the internal composition of the plot coincidentally.

An Opportunity

During a greeting with a friend, he didn't seem to be in a very good state, and I learned that he had been suffering from **the noise of the highway** next to his home and suffered from **frequent insomnia**. This has aroused my interest and thoughts, and I hope to be able to help.

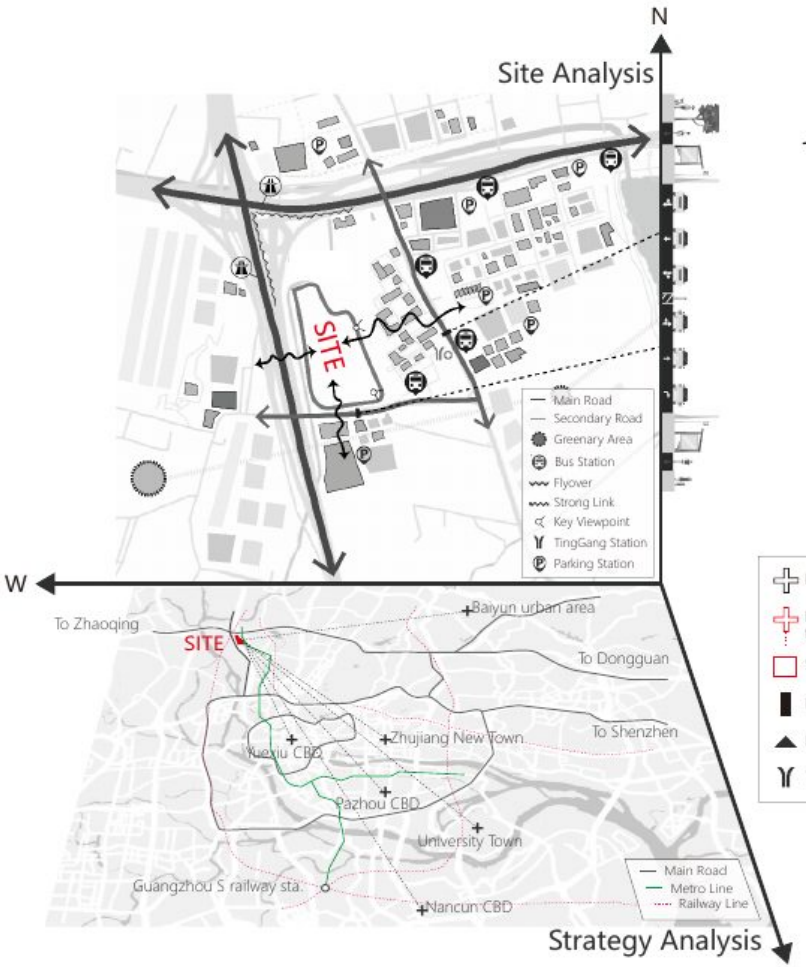


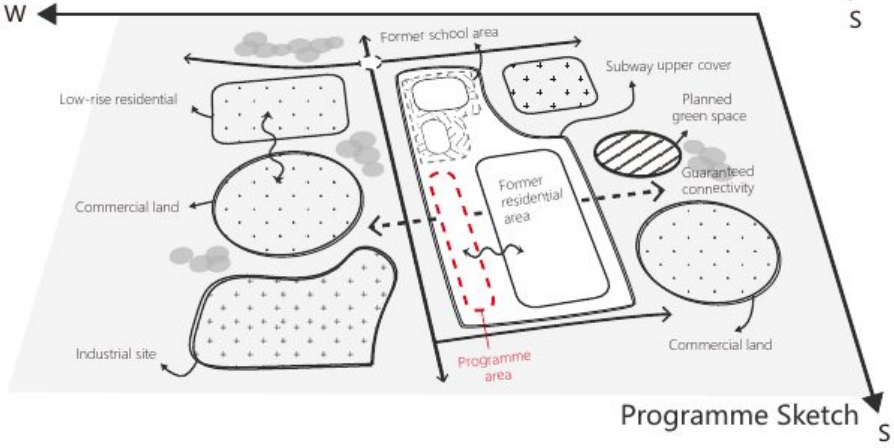
On-site Noise Survey Data

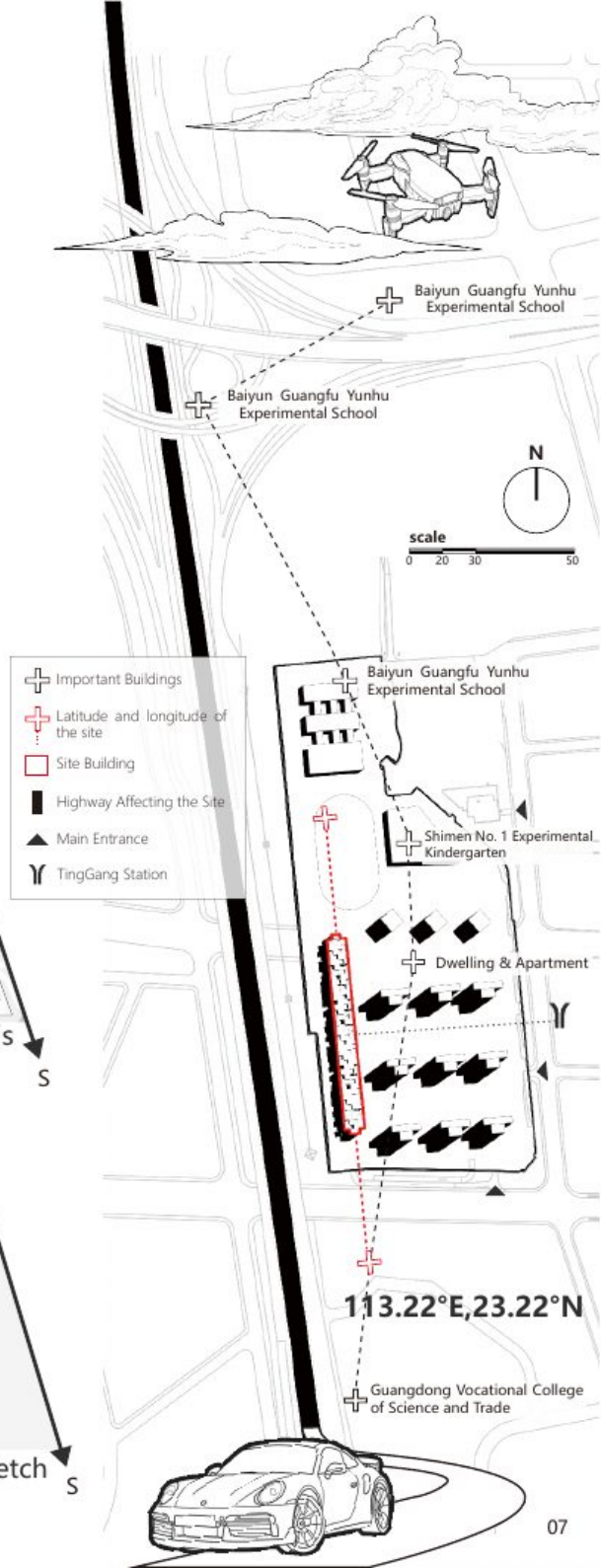


According to the investigation, it was preliminarily judged that the noise in the **low-rise residential** and **school areas** seriously exceeded the standard value, which had a great impact on daily use.

Preliminary Research







Logical Deduction

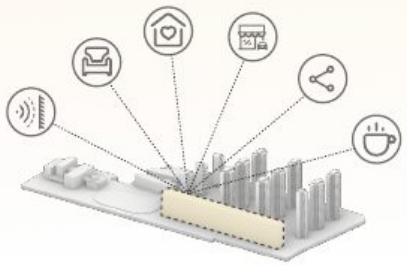
derive spatial logic based on community needs

Vertical exchange space
Breaking up the floor slabs between floors to form a joint vertical green space.

Semi-outdoor space
Open up the view between the floors, place greenery, create a courtyard, and improve the suitability of living.

Core space
Vertical traffic connects these combined block spaces to improve accessibility.

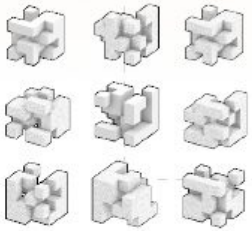
Aggregate Space
Blocks gathered together can form a large shared space, breaking down the barriers between layers.



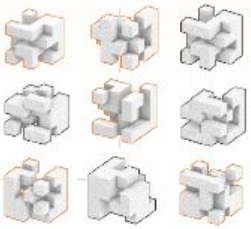
Objectives & Actions

The site needed to create a **mixed-use** building that not only solves the problem of sound insulation, but also meets the needs of commercial, apartment, community service, and shared leisure functions.

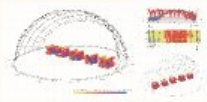
1. Randomly generate the 9x9 boxes.



2. Evaluate the possibilities of each square and select possible boxes.



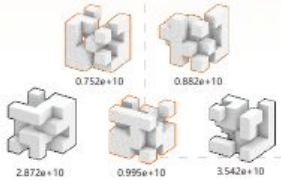
3. Evaluate its insolation conditions,Pick the top 5 boxes.



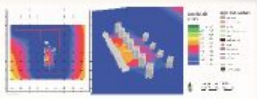
4. Verify the results of the assessment.



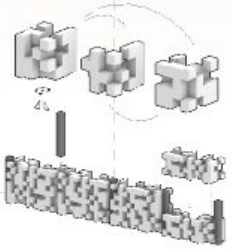
5. Extract the underside that mainly produces the echo.



6. calculate the area sum,Pick the top 3 boxes.



8. Noise evaluation of the results.



7. Permutations and combinations to form a staircase.





Derived from the isolation of noise, this conglomerate combining **commercial, residential, leisure, community services**, children's activities and other functional needs will become an **ideal utopia** for people in a high-density city.



03 Thriving Lingnan village

Combination of Traditional Lingnan Architectural Elements and Science & Technology Museums

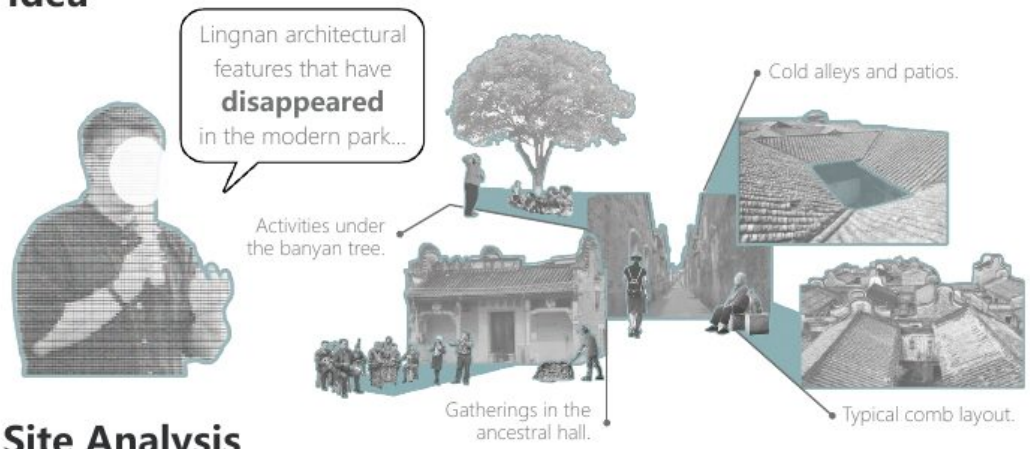
Location: Zengcheng District, Guangzhou, China
Date: Jul 2023 - Aug 2023
Instructor: Dr. Fengdeng Wan, Dr. Mingbo Xi
Acadamic Work: College Recommended Competitions, Group Work

The project participated in the **2024 International Solar Building Design Competition** on the topic of designing Zengcheng Science & Technology Museum.

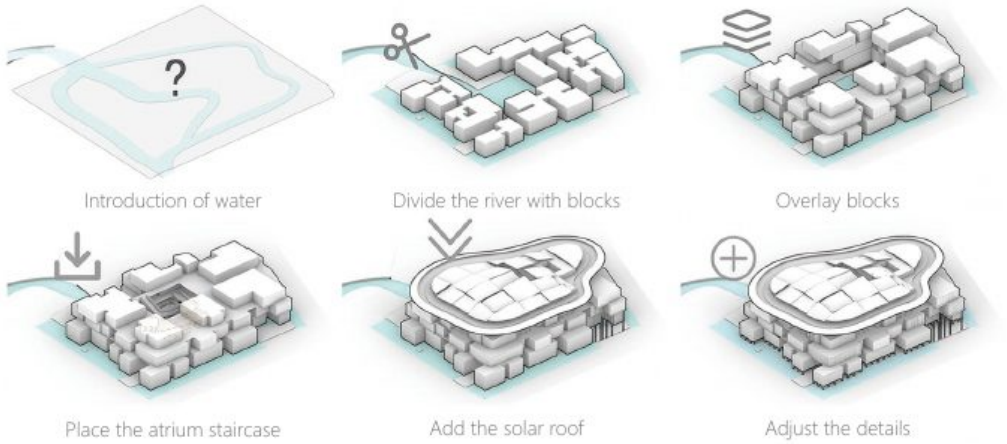
I was in charge of the preliminary analysis, the development of the program, the green and energy-saving design and carbon calculations, floor plans, detail drawings, analytical drawings, and the production of renderings.



Idea

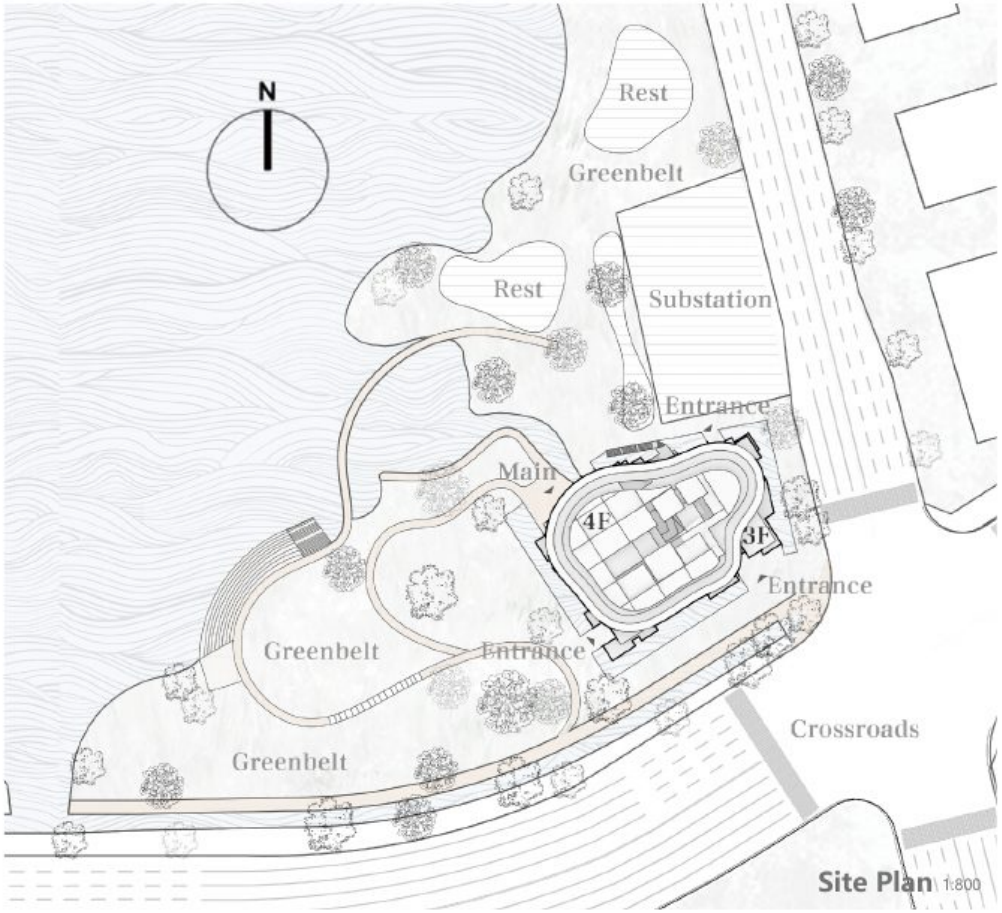


Block Generation



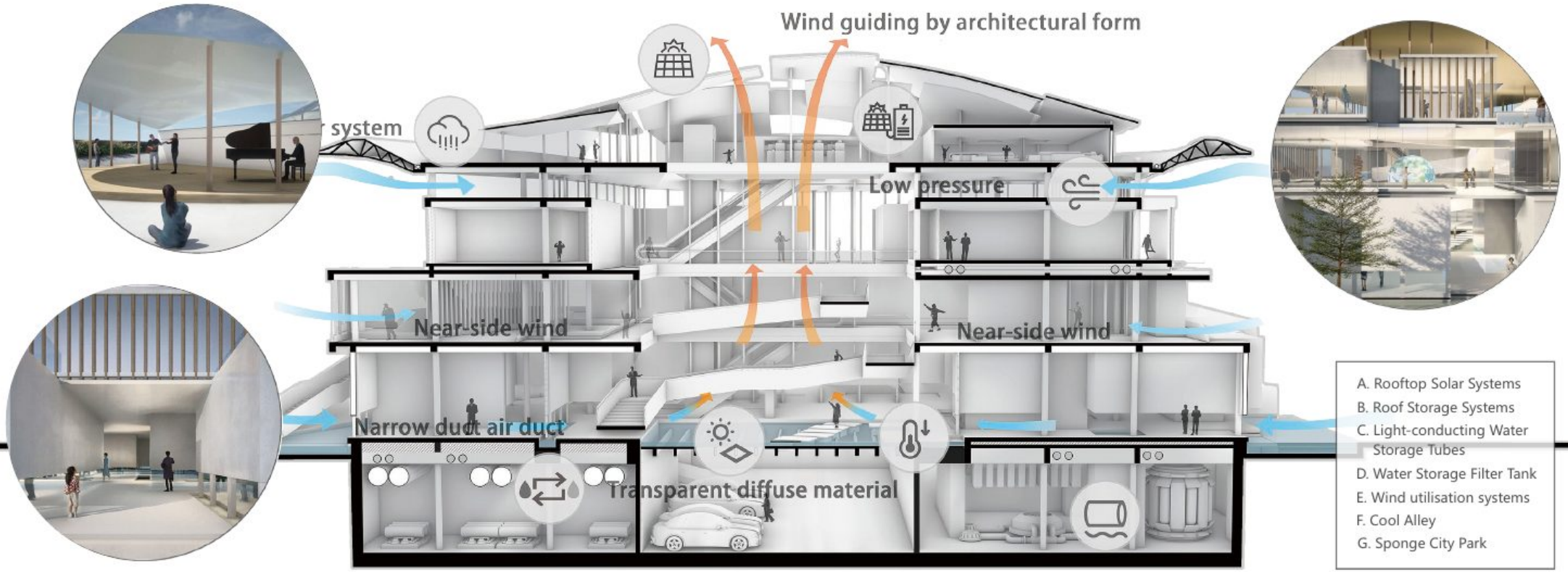
Site Analysis

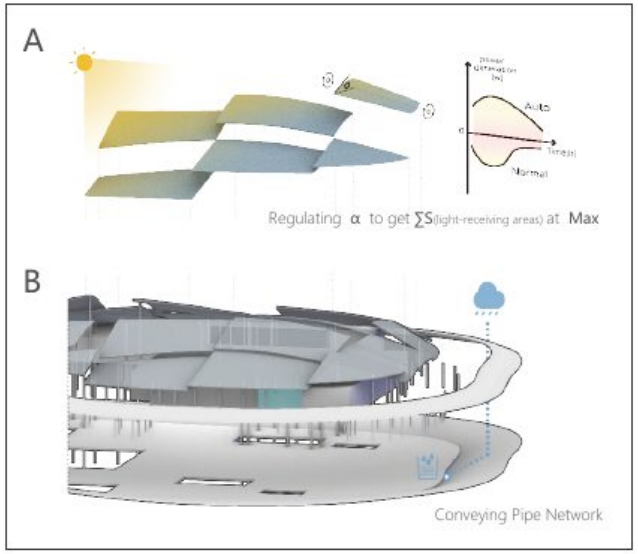


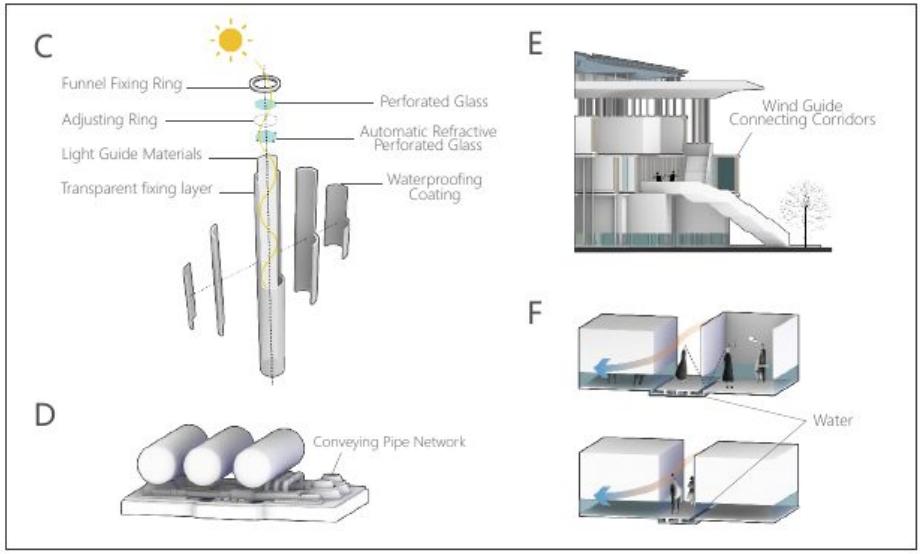


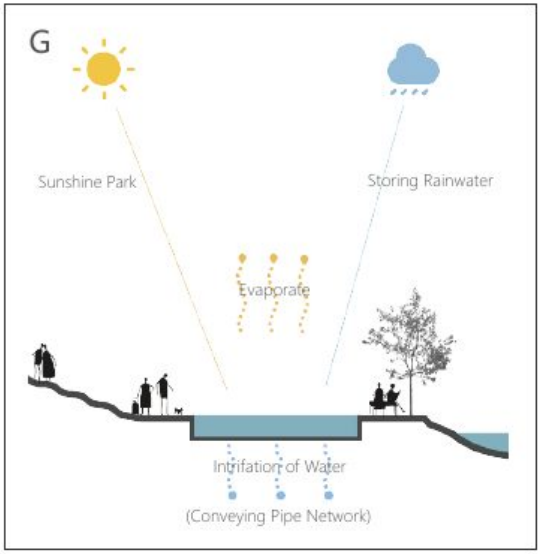
Energy-saving System

detailed introduction of nodes on water recycling and sunshine energy utilization



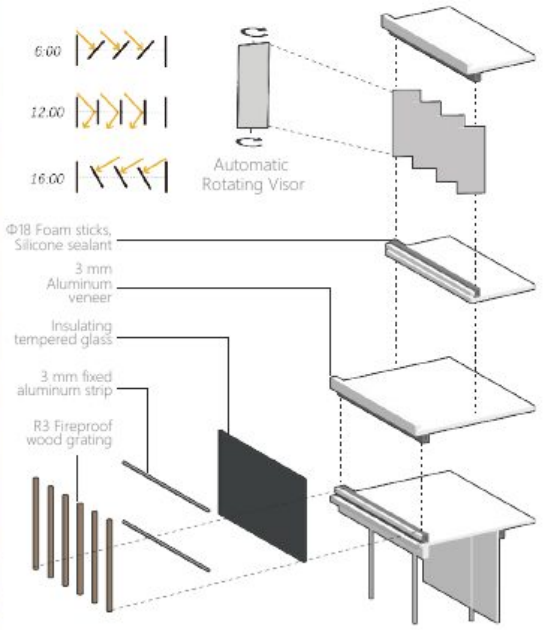




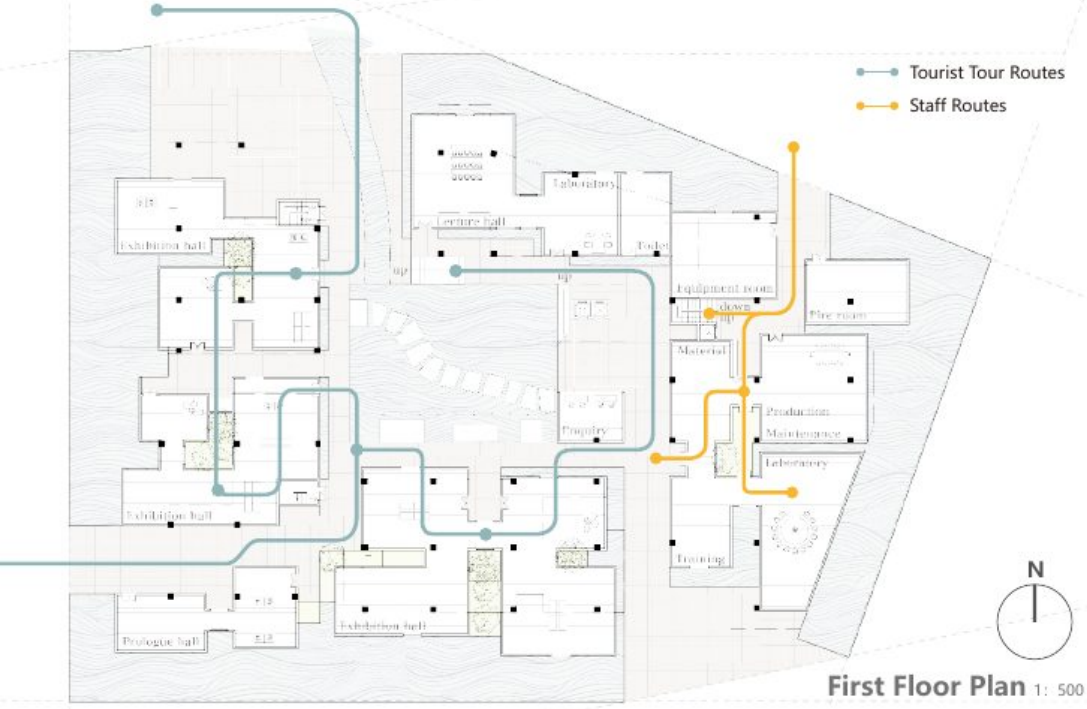




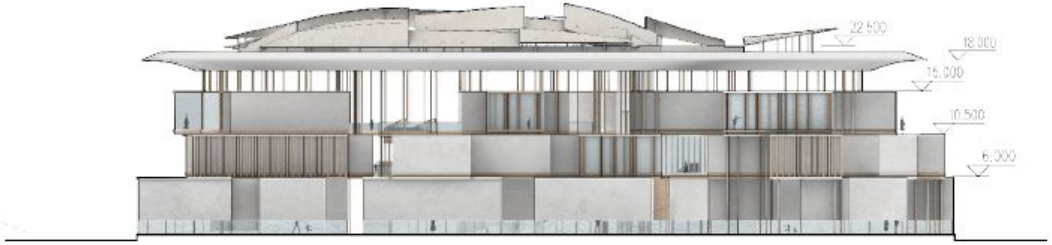




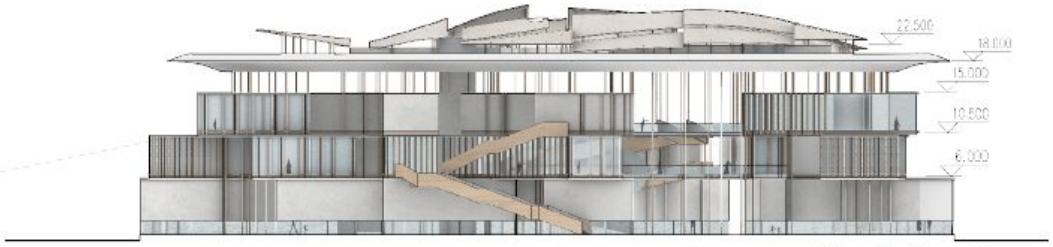
Glass Wall Details



First Floor Plan 1: 500



South Facade 1:500



North Facade 1:500



In southern mist and rain, the sky sinks low, Beneath thick shade, the banyan trees grow.

The pond's surface stirs with a gentle sway, By the alley's mouth, white gulls find their way.

- Southern Water Village at Dusk -

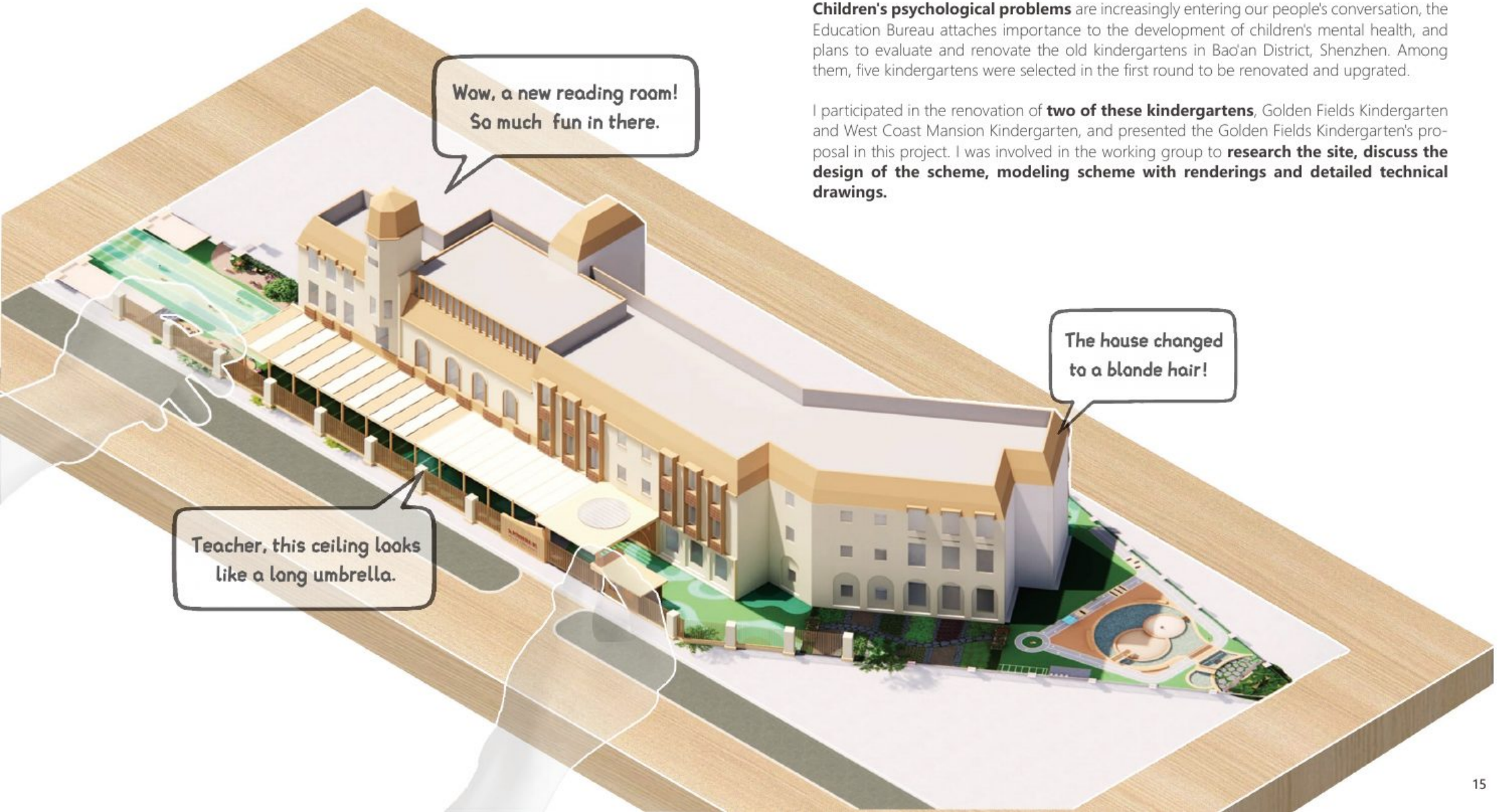
04 Smarten Up Kindergarten

Kindergarten Renovation Design based on the Perspective of Child Psychology

Location: Baoan District, Shenzhen, China
Date: Apr, 2024 - Jul, 2024
Internship work, Group Work

Children's psychological problems are increasingly entering our people's conversation, the Education Bureau attaches importance to the development of children's mental health, and plans to evaluate and renovate the old kindergartens in Bao'an District, Shenzhen. Among them, five kindergartens were selected in the first round to be renovated and upgrated.

I participated in the renovation of **two of these kindergartens**, Golden Fields Kindergarten and West Coast Mansion Kindergarten, and presented the Golden Fields Kindergarten's proposal in this project. I was involved in the working group to **research the site, discuss the design of the scheme, modeling scheme with renderings and detailed technical drawings.**



Preliminary Research



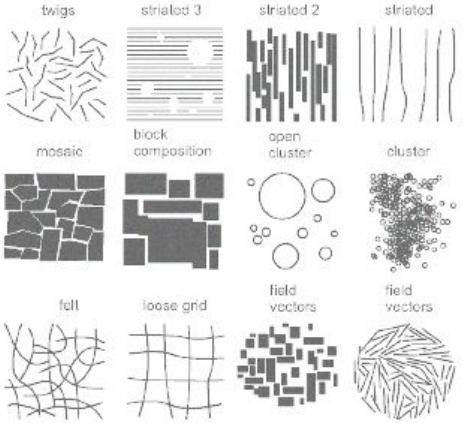
Children's Mental Health

Children's mental health has always been the focus of social concern. According to reports, at least 30 million children and adolescents in China suffer from mental illness. On March 1, 2021, the Institute of Psychology of the Chinese Academy of Sciences released the "Report on the Development of National Mental Health in China (2019-2020)", which showed that **in 2020, the detection rate of depression in primary schools was nearly 10%, and the detection rate of depression in adolescents was as high as 24.6%.**

Family & Child Brain Structure

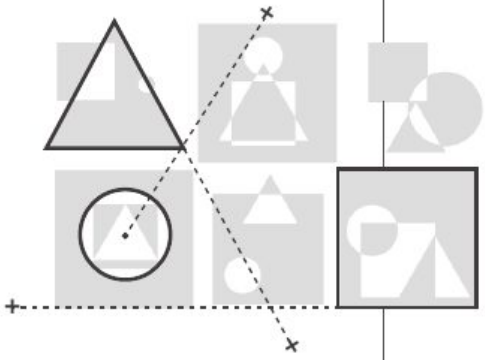
Recently, an international team composed of Cheng Wei, a young researcher from the Key Laboratory of Computational Neuroscience and Brain-like Intelligence of the Ministry of Education, Fudan University, and researchers from the University of Oxford and the University of Warwick in United Kingdom, uncovered the interaction between the family environment (family conflict and parental care) and children's brain structure, cognitive ability and mental health in a large-scale population for the first time. **The study found that the family environment was significantly associated with children's extroverted behavior problems (such as impulsive behavior), and it was also closely related to brain structure indicators in the orbitofrontal cortex, middle temporal gyrus, cingulate gyrus, and prefrontal lobe.**

Design Concept



Material

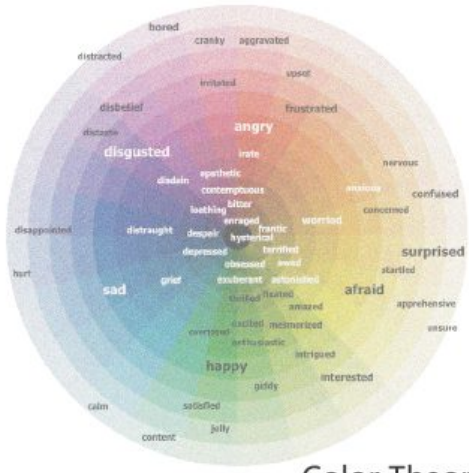
The sense of touch is a direct sensory pathway. Children perceive the texture of different materials through tactile touch, which transmits an abstracted emotional experience to the psyche, resulting in a healing effect.



Shapes

Human emotions are linked to certain shapes.

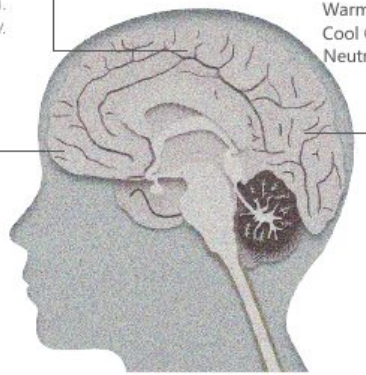
Circles/Ovals: positive, unity, love.
Triangles/Squares: balance, stability.
Vertical Lines: aggression, strength.
Horizontal Lines: calm, community.



Color Theory

Colors influence emotions & create connections.

Warm Colors: safety, positive, passionatehappy.
Cool Colors: cold, reserved, relaxed, proffesional.
Neutral Colors: critical, sophisticated, backdrops.



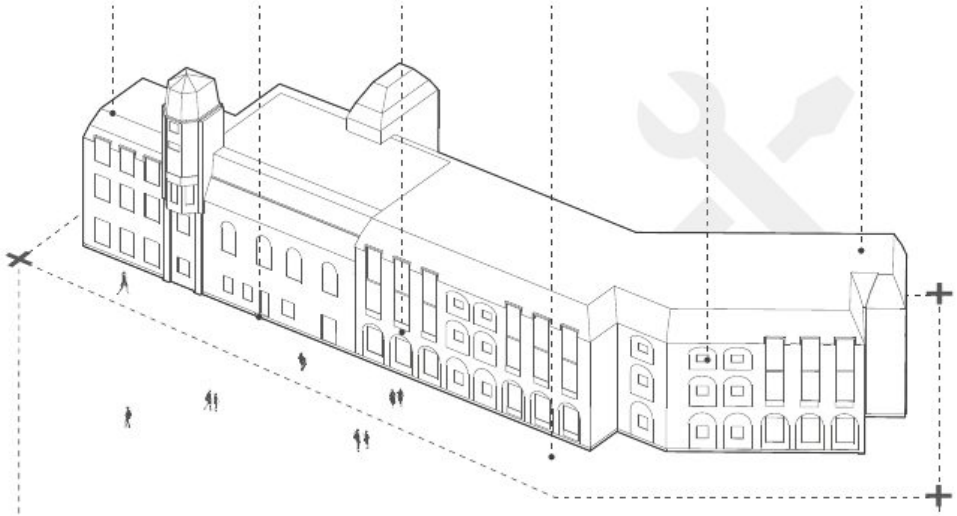


Site Analysis

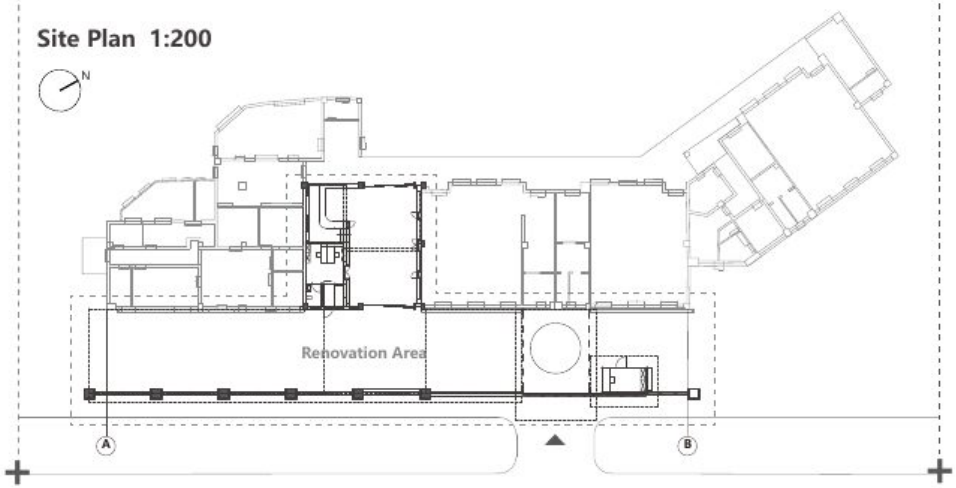
This project is a renovation program for **five** kindergartens located in Bao'an District of Shenzhen by the Shenzhen Municipal Bureau of Education, which requires the designers to design according to the current situation of the site, the needs of the kindergarten, and the direction of sustainable development. **One of them (Golden Field Kindergarten) is excerpted here.**



The kitchen exhaust pipe was reroofed | Renovation of the corridor system | Hall/health/reading room renovation | Restoration of the entrance gate | Facade elevation | Roof renovation (waterproofing & leak repair)



Site Plan 1:200



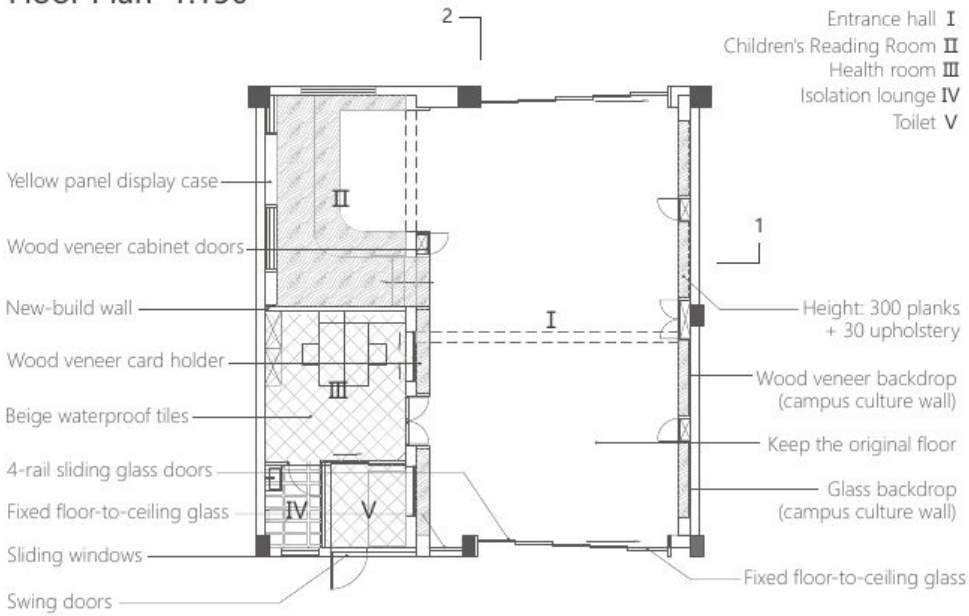




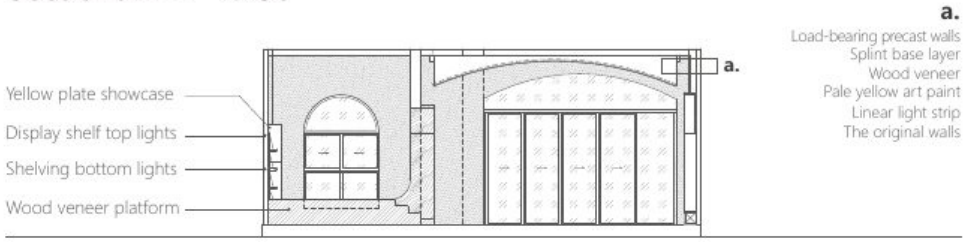
The interior follows the original curved elements for the window openings and ceiling lighting, and **the gold and yellow color scheme echoes the name and theme of the nursery**. A new semi-open reading room with movable furniture was created to give the children a sense of warmth and nature.

Renovation Part I, Indoor Halls

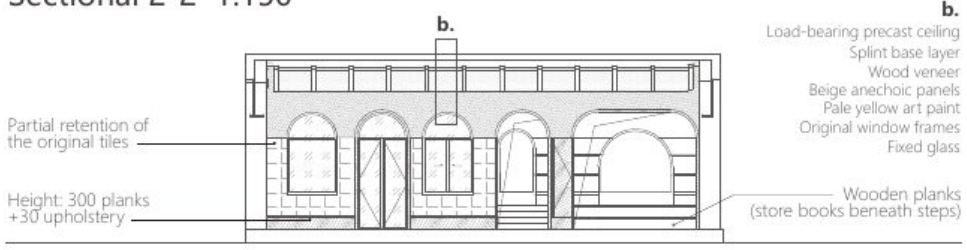
Floor Plan 1:150

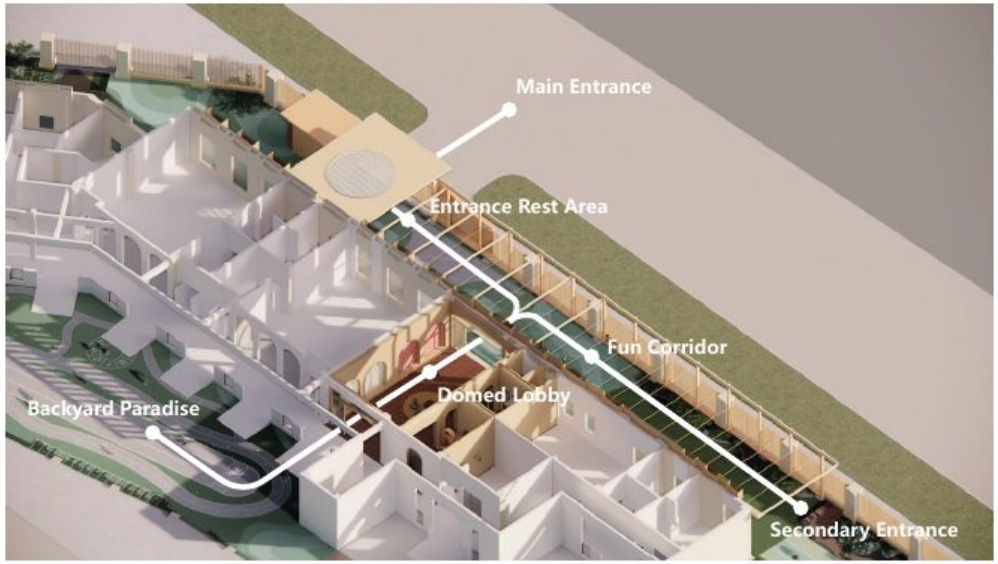


Sectional 1-1 1:150



Sectional 2-2 1:150







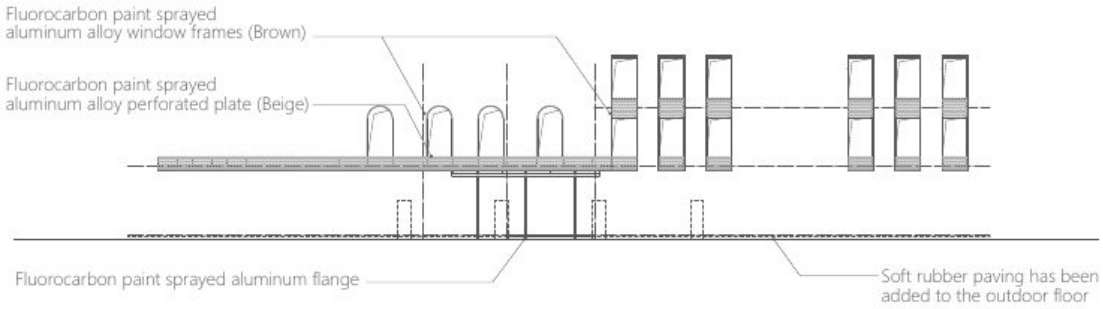




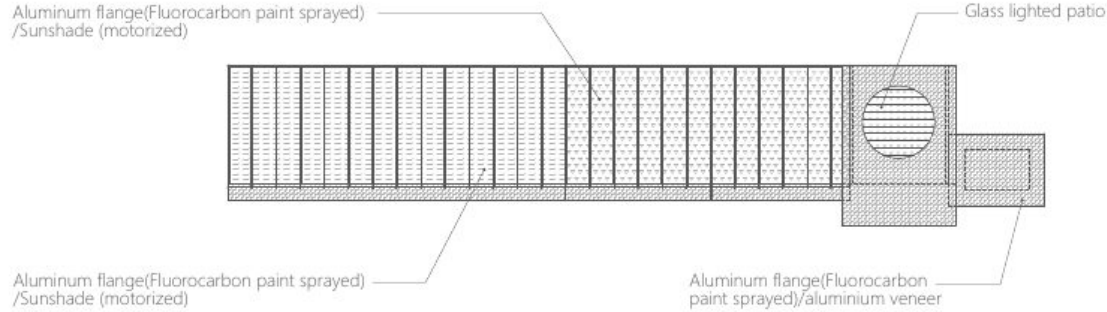
The overall outdoor promenade adopts a frame construction system, utilizing *sunshine panels outside the hall to solve the problem of **wind and rain protection**, and sunshade cloths in other parts to solve the problem of excessive **sunlight** for children's outdoor activities.

Renovation Part Ⅱ, Outdoor Space

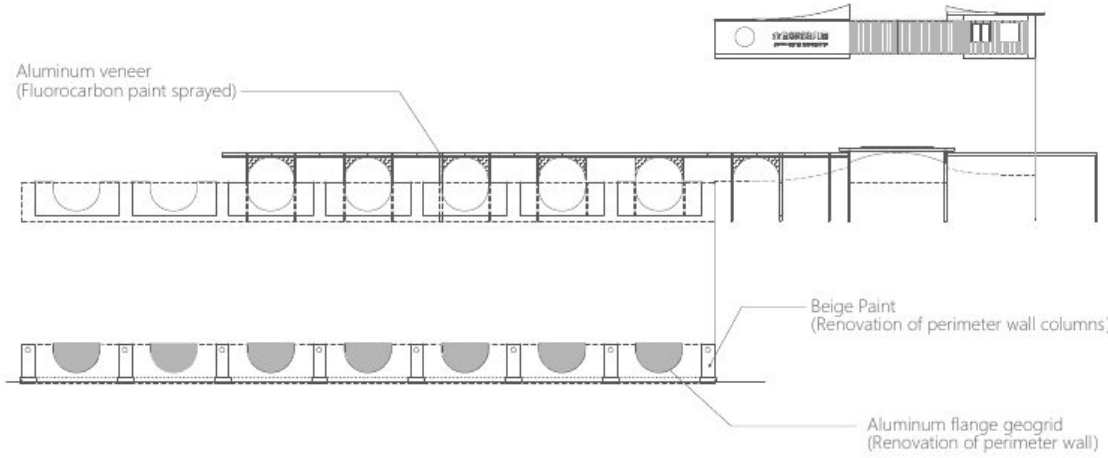
A-B Facades 1:150



Corridor Roof Plan 1:150



Corridor Facades 1:150











Project Results

The project began construction in July 2024, and due to **time and cost light factors**, the program is being constructed in **two phases**, with Phase 1 being completed in September 2024 with construction of the campus façade and connecting corridor improvements. The image to the left was taken on October 12, 2024 at the site.

We returned to the venue and made a **return visit to the garden teacher**. The garden was very satisfied with the current status of the project, and also **put forward a few problems and comments for the second phase** of the project for reference and improvement, such as the layout of the toilets and the intensity of the indoor lighting, etc.



Follow-up Works

Before the second phase of the project, we plan to **promote this project**. According to the dual perspective of education and design, we will organize small forums or salons for everyone to share the past and look forward to the future. The direction of our efforts, in addition to the mainstream architectural media such as Fanggu De, perhaps the Public Channel like **Bao'an Education, Shenzhen Education**, are also another platform that can be seen by more people (especially parents)!





05 Other Works

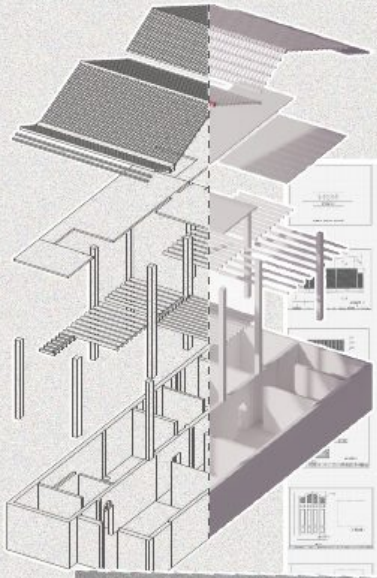


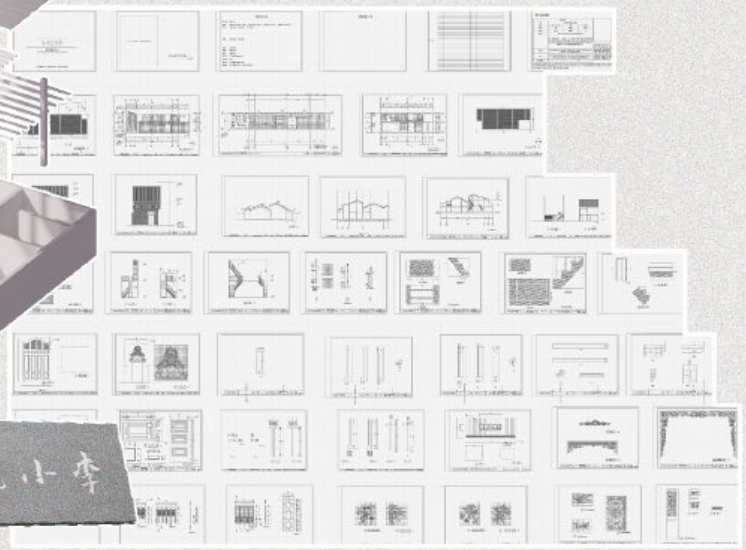


Ⅰ Analyze Church of Water, Tadao Ando

—Course: Art Sketching Practice 1,
Spring Term 2020-2021, individual work

Selected as the outstanding model of the college.



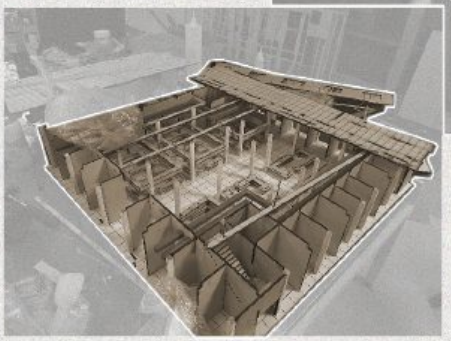


Ⅱ Mapping Bruce Lee's former residence

—Course: Survey and Drawing Practice,
Spring Term 2022-2023, group work

Responsible for plan, elevation, and detail sample surveying and modeling.







Ⅲ Boat & Market

—Model Making Week,
Fall Term 2022-2023, group work

Responsible for drawing design and model printing production.







Ⅳ Mountains, Water, Cities

—Paper Building Competition,
Nov 23,2021, group work

Responsible for conceptualization and paper modeling, poster design.

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